

Iberoamerican Mathematical Olympiad 2026

Friday, September 25, 2026

Problems from the 41st Iberoamerican Mathematical Olympiad (Buenos Aires, 2026), translated from Spanish.

Toolkit

- ◇ **Partial sums and residues.** For a sequence (a_n) , write

$$S_n = a_1 + \cdots + a_n$$

for its partial sums. Modulo m , the recurrence $S_n = S_{n-1} + a_n$ has only m possible states.

- ◇ **Term from sums and averages.** Note that

$$S_n = S_{n-1} + a_n \Rightarrow S_n - S_{n-1} = a_n.$$

If $n \mid S_n$, call $b_n = S_n/n$. Then

$$a_n = nb_n - (n-1)b_{n-1} = b_{n-1} + n(b_n - b_{n-1}) \quad (n \geq 2).$$

- ◇ **Polynomial divisibility.** If $p \in \mathbb{Z}[x]$ and $u, v \in \mathbb{Z}$, then

$$u - v \mid p(u) - p(v).$$

Also, if $1 = d_1 < \cdots < d_k = n$ are all positive divisors of n , then $d_i d_{k+1-i} = n$.

Definition. A partial order on a set A is a binary relation $\leq$ which is

- (a) **Reflexive.** For any $a \in A$, $a \leq a$.
- (b) **Antisymmetric.** For $a, b \in A$, $a \leq b$ and $b \leq a$ imply $a = b$.
- (c) **Transitive.** For $a, b, c \in A$, if $a \leq b$ and $b \leq c$, then $a \leq c$.

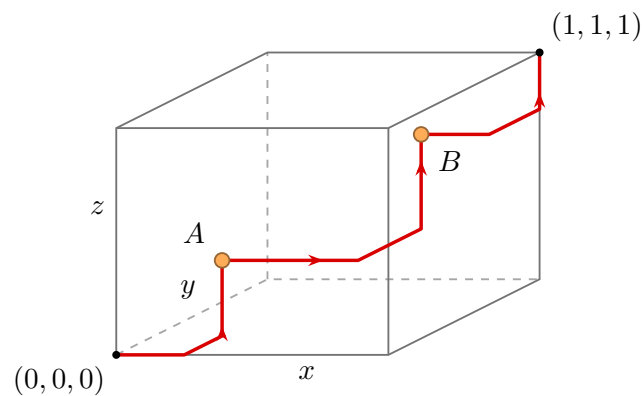
Example. For a set A , its power set along with the containment relation form a partial order $(\mathcal{P}(A), \subseteq)$.

Definition. In a poset (partially ordered set), a chain is totally ordered subset. This is, every element is comparable.

On the other hand, an antichain is a set of mutually incomparable elements.

Theorem (Dilworth). *In a finite poset, the maximum size of an antichain equals the minimum number of chains needed to cover all elements. This number is called the width of the partial order.*

Help for Problem 3



The red path collects alfajores located at $A = (\frac{1}{4}, \frac{1}{4}, \frac{1}{3})$ and $B = (\frac{3}{4}, \frac{2}{3}, \frac{3}{4})$. Every segment is parallel to a cube edge, and no coordinate decreases.



Problems: Day 1

Exercise 1. Let $a_1, a_2, \dots, a_{1000}$ be a permutation of the integers $1, 2, \dots, 1000$. Determine all possible values of a_1 for which, regardless of how the remaining integers are ordered, there is an integer k , $1 \leq k \leq 1000$, such that

$$a_1 + a_2 + \dots + a_k$$

is divisible by 3.

Exercise 2. Does there exist a bijection $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ satisfying all of the following conditions?

- (i) For every positive integer n , the sum $f(1) + f(2) + \dots + f(n)$ is divisible by n .
- (ii) For every positive integer n ,

$$f(1) + f(2) + \dots + f(n) \leq n^2.$$

- (iii) For infinitely many positive integers n ,

$$f(1) + f(2) + \dots + f(n) = n^2.$$

Note: Bijectivity means that each positive integer appears exactly once in the list $f(1), f(2), \dots$

Exercise 3. Mafalda dreams of taking an interstellar trip with her friends. On the journey, she discovers a cube-shaped room containing 2026 motionless *alfajores* floating inside it. She chooses a vertex of the cube and places all her accompanying friends there.

Each friend must travel to the opposite vertex, moving only in directions parallel to the edges of the cube and always away from the starting vertex, collecting every alfajor encountered along the way.

What is the minimum number of friends Mafalda must bring to guarantee that they can collect all the alfajores, regardless of their arrangement?

Note: Treat the friends and the alfajores as points in space.

Problems: Day 2

Exercise 4. A sequence $a_1, a_2, \dots$ of positive integers is called *porteña* if it satisfies the following conditions:

- (i) $a_1 < a_2 < a_3 < \dots$.
- (ii) $2a_n - n$ is a perfect square for every $n \geq 1$.
- (iii) $a_{n-1} + a_n \leq n^2$ for every $n \geq 2$.

Determine all *porteña* sequences.

Exercise 5. Let ABC be an acute triangle with $AB > AC$, inscribed in a circle Γ . A point X , distinct from A and C , varies along the arc AC of Γ that does not contain B . The lines AX and BC meet at P .

The line through P parallel to AB meets the tangent to Γ at B at D . The line through P parallel to AC meets the tangent to Γ at C at E .

Prove that the circumcenter of triangle XDE lies on a fixed line.

Note: The circumcenter of a triangle is the center of the circle through its three vertices.

Exercise 6. Determine all integers $n > 1$ for which there exist a permutation $d_1, d_2, \dots, d_k$ of all the positive divisors of n and a polynomial $P(x) \in \mathbb{Z}[x]$ such that

$$P(d_j) = d_{j+1} \quad (1 \leq j \leq k-1).$$

Solutions

Exercise 1. We may first consider the case for a_i to be a permutation of 4. Look at all the possible permutations and let us note which ones have a prefix sum divisible by 3.

FIGURE

From this, we see that the sums starting with 2 or 3 work. For the 1 and 4 we cannot eventually find a partial sum which is divisible by 3. In this fashion, we may conjecture:

A partial sum $a_1 + \cdots + a_k$ can be made divisible by 3 iff the starting number a_1 is either 0 or 2 mod 3.

Immediately if $a_1 \equiv 0 \pmod{3}$, the first partial sum a_1 is divisible by 3. We are left with analyzing:

If $a_1 \equiv 1 \pmod{3}$, then in the remaining 999 elements, every residue mod 3 is equidistributed. Therefore, mod 3 we can add 1, then 2 the same number

Exercise 2 (Integer averages and equality cases). **No such bijection exists.** Write $S_n = \sum_{j=1}^n f(j) = nb_n$, where b_n is an integer with $1 \leq b_n \leq n$. If $b_n < b_{n-1}$, then

$$f(n) = nb_n - (n-1)b_{n-1} \leq b_{n-1} - n \leq -1,$$

a contradiction. Therefore (b_n) is nondecreasing.

Suppose $S_N = N^2$, so that $b_N = N$. For every $m > N$,

$$f(m) = b_{m-1} + m(b_m - b_{m-1}) \geq N.$$

Surjectivity forces $1, \dots, N-1$ to occur among the first N values. Injectivity and their sum then give

$$\{f(1), \dots, f(N)\} = \left\{1, \dots, N-1, \frac{N(N+1)}{2}\right\}.$$

We show that $N \geq 5$ is impossible. Since $f(1) = 1$ and

$$f(N) = N^2 - S_{N-1} \equiv 1 \pmod{N-1},$$

the value $f(N)$ cannot be among $1, \dots, N-1$: the only candidate is the already used value 1. Hence

$$f(N) = \frac{N(N+1)}{2}, \quad S_{N-1} = \frac{N(N-1)}{2}.$$

Divisibility by $N-1$ makes N even, hence $N \geq 6$.

Now $f(N-1)$ lies in $\{1, \dots, N-1\}$, and

$$f(N-1) \equiv S_{N-1} \equiv N/2 \pmod{N-2}.$$

For $N \geq 6$, the only possible value is $N/2$. Consequently $S_{N-2} = N(N-2)/2$, and similarly

$$f(N-2) \equiv S_{N-2} \equiv N/2 \pmod{N-3}.$$

Again the only representative in $\{1, \dots, N-1\}$ is $N/2$, because $N/2 - (N-3) \leq 0$ and $N/2 + (N-3) > N-1$. This contradicts injectivity. Equality can occur only when $N \leq 4$, so it cannot occur infinitely often.

Exercise 3 (Monotone paths and partial orders). The minimum number is 1519.

For a fixed starting vertex, choose coordinates so that the cube is $[0, 1]^3$ and that vertex is the origin. A single friend can visit a list of points precisely when they form a chain for coordinatewise order:

$$(x, y, z) \leq (x', y', z') \iff x \leq x', y \leq y', z \leq z'.$$

Indeed, the coordinates along a permitted path never decrease, and any chain can be joined by segments parallel to the cube's edges. By Dilworth's theorem, the number of friends needed equals the largest antichain size in this order.

Opposite starting vertices reverse the order and have the same chains and antichains. Thus there are four relevant orders, represented by the coordinate directions

$$v_1 = (1, 1, 1), \quad v_2 = (1, 1, -1), \quad v_3 = (1, -1, 1), \quad v_4 = (1, -1, -1).$$

In order i , compare coordinates after multiplying them by the corresponding entries of v_i . Every pair of distinct points is comparable in at least one of these orders: choose signs matching their coordinate differences, up to reversing all three signs.

Upper bound. If all four orders had antichains A_i of size 1520, their total number of memberships would be $4 \cdot 1520 = 6080$. But no two distinct points can belong to all four antichains, since they would be incomparable in every order. Thus at most one point belongs to four of the sets, and each other point belongs to at most three. This gives at most

$$4 + 3 \cdot 2025 = 6079$$

memberships, a contradiction. Some order therefore has width at most 1519, and Mafalda can choose its starting vertex.

Lower bound. Center the cube at the origin. Place respectively 507, 507, 506, 506 points on the four rays tv_i , $t > 0$. Choose all 2026 parameters t distinct and sufficiently small that the points lie in the cube's interior.

For two points sv_i, tv_j with $s > t$, every coordinate of their difference has the sign of the corresponding coordinate of v_i : its absolute value is either $s - t$ or $s + t$. Thus they are comparable in exactly order i . In particular, in order i , all points outside ray i form an antichain. Each such antichain has at least $2026 - 507 = 1519$ points. Every choice of starting vertex therefore needs at least 1519 friends, proving the claim.

Exercise 4 (A sequence squeezed between two bounds). The unique sequence is

$$a_n = n(n+1)/2. \text{ Write}$$

$$2a_n - n = b_n^2, \quad b_n \in \mathbb{Z}_{\geq 0}.$$

Since the positive integers a_n strictly increase,

$$b_{n+1}^2 - b_n^2 = 2(a_{n+1} - a_n) - 1 \geq 1.$$

Thus the nonnegative integers b_n strictly increase. Also $b_1^2 = 2a_1 - 1 \geq 1$, so $b_n \geq n$ for all n . On the other hand, the last condition gives

$$b_{n-1}^2 + b_n^2 \leq 2n^2 - 2n + 1 = (n-1)^2 + n^2.$$

The lower bounds force equality, hence $b_n = n$ for every n . Therefore $a_n = (n + n^2)/2$. Conversely, these triangular numbers strictly increase, satisfy $2a_n - n = n^2$, and have $a_{n-1} + a_n = n^2$.

Exercise 5 (Coordinates along the sides of the triangle). Use $\overrightarrow{AB}, \overrightarrow{AC}$ as a coordinate basis, so that $A = (0, 0)$, $B = (1, 0)$, and $C = (0, 1)$. Put

$$r = AB^2, \quad s = AC^2, \quad h = 2\overrightarrow{AB} \cdot \overrightarrow{AC}, \quad R = r + s, \quad K = R - h = BC^2.$$

Squared distance from A is the positive definite quadratic form

$$Q(x, y) = rx^2 + hxy + sy^2.$$

In particular, a Euclidean circle has equation $Q(x, y) + Ux + Vy + W = 0$. The original circle is $\Gamma : Q(x, y) - rx - sy = 0$, and its tangents at B, C are

$$rx + (h - s)y = r, \quad (h - r)x + sy = s.$$

Describe the moving points. Write $P = (a, b)$, where $a = 1 - t$ and $b = t$. The parallel-line conditions and the tangent equations give

$$D = \left(a + \frac{bK}{r}, b\right), \quad E = \left(a, b + \frac{aK}{s}\right).$$

Since A, X, P are collinear, substituting $X = \lambda(a, b)$ in Γ gives

$$q = Q(a, b), \quad m = ra + sb, \quad X = \frac{m}{q}(a, b).$$

Here $q > 0$. Also $a, b \neq 0$: otherwise AX would be AC or AB , making $X = C$ or $X = B$, which the prescribed arc excludes.

Find the circle through X, D, E . The useful calculation is to write its equation so that its parameter dependence is visible. Consider

$$\begin{aligned} F_t(x, y) &= h(Q(x, y) - rx - sy) \\ &\quad + ra(R(1 - x) - hy) + sb(R(1 - y) - hx). \end{aligned}$$

Because $h > 0$, this has the quadratic part of a circle. We verify that $F_t = 0$ passes through the required points.

At $X = \lambda(a, b)$, the first term vanishes, and

$$\begin{aligned} F_t(X) &= Rm - \lambda(Rra^2 + hab(r + s) + Rsb^2) \\ &= Rm - \lambda Rq = 0. \end{aligned}$$

For $D = (D_x, D_y)$, direct substitution using $a + b = 1$ gives

$$\begin{aligned} Q(D) - rD_x - sD_y &= \frac{sb^2K}{r}, \\ R(1 - D_x) - hD_y &= -\frac{sbK}{r}, \\ R(1 - D_y) - hD_x &= K\left(a - \frac{hb}{r}\right). \end{aligned}$$

Therefore

$$F_t(D) = \frac{hsb^2K}{r} - asbK + sbK \left(a - \frac{hb}{r} \right) = 0.$$

Interchanging (r, a, x) with (s, b, y) gives $F_t(E) = 0$. The points are noncollinear; indeed,

$$\det(D - X, E - X) = \frac{hK^2a^2b^2}{rsq} > 0.$$

Thus $F_t = 0$ is the circumcircle of XDE .

Read off the center. Write $F_t = hQ + L(t)x + M(t)y + N(t)$. All three coefficients are affine in t , because $a = 1 - t$ and $b = t$. The center (u_t, v_t) satisfies

$$\begin{pmatrix} 2r & h \\ h & 2s \end{pmatrix} \begin{pmatrix} u_t \\ v_t \end{pmatrix} = -\frac{1}{h} \begin{pmatrix} L(t) \\ M(t) \end{pmatrix}.$$

The fixed matrix is invertible, since $4rs - h^2 = 4AB^2AC^2 \sin^2 \angle BAC > 0$. Hence $(u_t, v_t) = (u_0, v_0) + t(u_1, v_1)$ for fixed vectors, so the circumcenter lies on a fixed line.

Exercise 6 (A polynomial orbit through the divisors). The answer is

$$n = p^a \quad \text{or} \quad n = p(2p - 1),$$

where p is prime, $a \geq 1$, and in the second family $2p - 1$ is also prime.

For an integer polynomial, $x - y \mid P(x) - P(y)$. Consequently, whenever $i, j < k$,

$$d_i - d_j \mid d_{i+1} - d_{j+1}. \quad (*)$$

The successors are distinct, and the list contains no fixed points or cycles: either would repeat a divisor.

Step 1: for composite n , the last divisor is n . Prime n already belongs to a claimed family, so assume that n is composite. Let p be its smallest prime factor and $M = n/p$ its largest proper divisor. Thus $k \geq 3$ and $M \leq n/2$.

If neither 1 nor n were last, then $n - 1 \mid P(n) - P(1)$. The images are distinct divisors of n , so their difference has absolute value at most $n - 1$. Hence their images must be 1, n , creating fixed points or a two-cycle. Therefore $d_k \in \{1, n\}$.

Let $g_j = |d_{j+1} - d_j|$. Relation (*) gives $g_j \mid g_{j+1}$, so the positive gaps are nondecreasing. Suppose $d_k = 1$. If n occurs before position $k - 1$, the gap following it is at least $n - M$, whereas the last gap is at most $M - 1 < n - M$, a contradiction. If $d_{k-1} = n$, put $x = d_{k-2}$. Then $1 < x \leq M$, and

$$n - x \mid n - 1, \quad \frac{n - 1}{2} < n - x < n - 1.$$

This is impossible, since a proper positive divisor of $n - 1$ is at most $(n - 1)/2$. Thus $d_k = n$.

Step 2: examine the images of 1 and M . If $n = p^2$, it is already a prime power. Otherwise $M > p$. Since $M - 1 \mid P(M) - P(1)$, their images form either the pair $\{1, M\}$ or the pair $\{p, n\}$. Indeed, two proper divisors lie between 1 and M , so a nonzero difference divisible by $M - 1$ forces the first pair. If one image is $n = pM$ and the other is $u \leq M$, then $u \equiv p \pmod{M - 1}$, forcing $u = p$ because $1 < p < M$. The first pair creates fixed points or a two-cycle. Therefore $\{P(1), P(M)\} = \{p, n\}$.

Case A: $P(1) = p$ and $P(M) = n$. For a proper divisor $x > 1$, $x - 1 \mid P(x) - p$. If $P(x) < p$, then $P(x) = 1$, since p is the smallest divisor above 1. This would give $x - 1 \mid p - 1$, hence $x = p$, producing the two-cycle $1 \leftrightarrow p$. Also $P(x) \neq p$ by injectivity on the proper divisors. Thus $P(x) > p$, so $P(x) - p \geq x - 1$ and $P(x) > x$. The whole list is therefore increasing. Write it as

$$1 = e_1 < e_2 = p < \cdots < e_{k-1} = M < e_k = n.$$

For $3 \leq j \leq k-1$, repeatedly apply (*) to the pair (e_2, e_j) , shifting both indices forward $k-1-j$ times:

$$e_j - p \mid e_{k-1} - e_{k+1-j} = \frac{n}{p} - \frac{n}{e_j} = \frac{n(e_j - p)}{pe_j}.$$

It follows that

$$pe_j \mid n \quad (3 \leq j \leq k-1). \quad (**)$$

Suppose n is not a power of p , and let p^a be its highest power of p . If $a \geq 2$, then p^a is a proper divisor different from $1, p$, so (**) would imply $p^{a+1} \mid n$. Thus $a = 1$, and $n = pM$ with $p \nmid M$. If M were composite, choose a divisor u with $1 < u < M$. Then pu is a proper divisor different from $1, p$, and (**) would give $p^2u \mid n$, again a contradiction. Hence $M = q$ is prime, with $q > p$.

The list is now $1, p, q, pq$. Its consecutive gaps give

$$p-1 \mid q-p, \quad q-p \mid q(p-1).$$

Writing $q-p = m(p-1)$ gives $m \mid q$. Since $1 \leq m < q$ and q is prime, $m = 1$, so $q = 2p-1$.

Case B: $P(1) = n$ and $P(M) = p$. Now $M-p \mid P(p)-p$, and $P(p) \neq p, n$ by injectivity. If $P(p) > p$, then $P(p) \leq M$, so divisibility forces $P(p) = M$, giving the two-cycle $p \leftrightarrow M$. Therefore $P(p) = 1$, so $M-p \mid p-1$ and $M \leq 2p-1$. Every prime factor of M is at least p . If M were composite, then $M \geq p^2 > 2p-1$. Hence $M = q$ is prime and $n = pq$, with the forced ordering $q, p, 1, pq$. Its gaps give $q-p \mid p-1$ and $p-1 \mid pq-1$. The second divisibility implies $p-1 \mid q-1$, hence $p-1 \mid q-p$. Thus $q-p = p-1$, and again $q = 2p-1$.

Step 3: construct the polynomial in each family. For $n = p^a$, use the ordering $1, p, \dots, p^a$ and $P(x) = px$. For $n = pq$, where $q = 2p-1$ is prime, use the ordering $1, p, q, pq$ and

$$P(x) = x^2 - px + q.$$

Indeed, $P(1) = p$, $P(p) = q$, and $P(q) = q(q-p+1) = pq$. Both families therefore work.